Report

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Introduction

The goal of this project is to implement a lane detection pipeline using Python and OpenCV.
The pipeline detects **white and yellow lanes** in road images and produces:

- The final image with detected lane overlays
- **I**ntermediate images after each major processing step for visualization

This approach is useful in autonomous driving and driver assistance systems.

Methodology

The lane detection pipeline consists of the following steps:

1. **HSV Color Masking**
    - Convert the input image to HSV color space.
    - Apply masks to isolate **yellow and white lanes**.
2. **Gaussian Blur**
    - Smooth the masked image to reduce noise.
    - Helps improve edge detection accuracy.

3. **Custom Canny Edge Detection**

   - Compute gradients using Sobel operators.
   - Apply **non-maximum suppression**, double threshold, and hysteresis to extract edges.

4. **Region of Interest (ROI) Masking**

    - Apply a trapezoidal mask to focus on the road area.
    - Reduces false positives from irrelevant regions.

5. **Hough Line Transform**

    - Detect straight line segments from the edge map.
    - Overlay detected lines on the original image for visualization.

6. **Lane Line Fitting**

    - Separate detected lines into **left** and **right lanes** based on slope.
    - Fit a single line for each lane using linear regression
    - Draw the final lane overlay on the image.

Results

- Gaussian blurred masks
- Edges detected by the custom Canny algorithm
- Edges limited to the road region
- Line segments detected by Hough Transform

Discussion

- The pipeline successfully detects lanes in **well-lit, clearly marked roads**.
- Challenges:
    - Faint or broken lane markings can reduce detection accuracy.
    - Shadows, rain, or night conditions require advanced preprocessing.
- Improvements:
    - Adaptive thresholding for varying lighting conditions.
    - Smoothing lane trajectories over multiple frames in video sequences.

Conclusion

- A robust, modular lane detection pipeline has been implemented.
- Outputs all intermediate steps for better visualization and debugging.

- The script is **command-line friendly** making it suitable for batch processing in terminal environments